ENGINEERING DESIGN REVIEW

TRM4-Inspired EO Performance Application

Consolidated changes to the interface, modeling engine, imports, outputs, and verification program.

IN SCOPE
Standard TRM-based DRI, calibrated imaging physics, atmosphere, camera measurements, usability.

EXCLUDED
New or adaptive Johnson criteria, observer-feedback learning, and learned DRI thresholds.



01 Executive decision

Build a compact, traceable electro-optical performance tool based on TRM4 principles, not a screen-by-screen replica of the original software. Retain essential physical models, accept measured data as an alternative to internal assumptions, and make the normal workflow simple.

01

USER EXPERIENCE

Quick / Engineering / Lab

→

02

PHYSICS ENGINE

Optics / Signal / Atmosphere

→

03

VALIDATION & DATA

Calibration / Imports / Tests

Four disposition rules

DECISION	MEANING
KEEP	Expose essential inputs and results in the main workflow.
SIMPLIFY	Calculate automatically, load from a sensor profile, or use a well-labeled preset.
ADVANCED	Retain in the engine or an expert panel only when it changes the result.
REMOVE	Omit from the normal product scope unless a real supported sensor requires it.

Top changes to implement

- Use three workflows: **Quick Prediction**, **Engineering Analysis**, and **Laboratory & Image Evaluation**.
- Make camera profiles the main input method, with FOV, IFOV, Nyquist and dependent values calculated automatically.
- Support external optical MTF and transmission imports with explicit data precedence and unit validation.
- Preserve measured noise/MTDP/MCP inputs, but hide rarely used sensor electronics and legacy video hardware.
- Offer a fast atmospheric model and a spectral atmospheric-data import route, plus path-radiance and turbulence options.
- Add sensitivity analysis, model diagnostics, laboratory validation, traceability and uncertainty reporting.

Critical distinction: An item marked REMOVE may still matter for a specialized imaging system. Removing it from this product scope is not the same as claiming that the physics is unnecessary.

02 Inputs: imager, optics & sampling

FUNCTION / INPUT	DECISION	IMPLEMENTATION RULE
Imager category / sensing band	KEEP	Visible/NIR, SWIR, MWIR, LWIR; choose an appropriate physics branch.
Imager type	SIMPLIFY	Default to staring digital; infer cooled/uncooled or CCD/CMOS from a profile.
Sensor resolution, pixel pitch	KEEP	Specify or load horizontal and vertical values; validate effective active area.
Focal length / optical mode	KEEP	Allow multiple zoom/focus profiles, not only a nominal value.
FOV, IFOV, Nyquist frequency	SIMPLIFY	Calculate and display; allow measured FOV as a cross-check.
Frame rate / video vs still	KEEP	Required for temporal integration and output assumptions.
F-number / clear aperture	SIMPLIFY	Use one principal entry and calculate its consistent counterpart.
Pupil geometry / obscuration	ADVANCED	Enable only for relevant optics; otherwise use a standard aperture model.
Diffraction MTF	KEEP	Calculate internally only when not already included in imported optics data.
Aberration/blur approximation	ADVANCED	Fallback when full optical MTF is unavailable; label approximation.
Measured/simulated optics MTF	KEEP	CSV import with frequency units, direction, spectrum and field metadata.
Optics transmission	KEEP	Scalar efficiency or full wavelength-dependent absolute curve.
Pixel binning, micro-scan/dither	ADVANCED	Enabled only for hardware that actually performs these operations.
Mechanical scan / analog interlace	REMOVE	Out of normal scope for staring digital day, SWIR and thermal cameras.
Two identical sensors, unusual FOV geometry	ADVANCED	Special-purpose capability, not a default form control.

TRM4 source: Chapter 2, Sections 2.2-2.6 (pp. 18-34); Chapter 3, Sections 3.2-3.4 (pp. 104-114); Chapter 4, Sections 4.1-4.2 (pp. 161-179).

03 Inputs: detector, electronics & display

FUNCTION / INPUT	DECISION	IMPLEMENTATION RULE
Spectral QE / responsivity	KEEP	Use detector-profile curves; require valid wavelength range and units.
Integration time / exposure	KEEP	Used in photon budget, noise and motion-blur calculations.
NETD for uncooled thermal	KEEP	Allow measured camera-level value at documented test conditions.
MTDP / MRTD / MCP / MRC	ADVANCED	Optional measured performance curves; replace corresponding modeled chain when valid.
Dark current, read noise, FWC	ADVANCED	Used in detailed charge-based sensitivity and saturation calculations.
Gain, ADC resolution, LSB	ADVANCED	Avoid as routine inputs unless validated measurements require them.
FPN / spatial non-uniformity	ADVANCED	Do not conflate spatial noise with temporal noise or NETD.
IETD / detector non-uniformity	ADVANCED	Support measured data; do not assume a universal fixed fraction of NETD.
D*, TDI, CTE and scanning details	ADVANCED	Only for detector architectures that need those models.
Digital filtering / frame averaging	SIMPLIFY	Choose actual processing path; measured output SFR may replace submodels.
Output compression / video processing	ADVANCED	Only when compression or temporal processing changes delivered imagery.
CRT, DAC, legacy video formats	REMOVE	Omit for modern fully digital pipelines, unless a supported configuration uses them.
Image intensifier and coupling	REMOVE	Only reintroduce for ICCD/ICMOS/NVG product support.
Display and eye parameters	SIMPLIFY	Standardized observation profile; advanced overrides for observer model.
Storage gate / detailed electronics	ADVANCED	Internal diagnostics rather than normal operator inputs.

TRM4 source: Sections 2.7-2.18 (pp. 34-76); Sections 3.5, 3.8-3.11 (pp. 114-152).

04 Inputs: targets, environment & performance

FUNCTION / INPUT	DECISION	IMPLEMENTATION RULE
Target dimensions / task	KEEP	Explicit height, width, view; use standard targets as selectable presets.
Target-background signature	KEEP	Temperature difference for thermal; reflectance/radiance contrast for reflected bands.
Background temperature/reflectance	KEEP	Visible in advanced scenario settings; critical for signal/noise estimates.
3D target orientation, altitude	ADVANCED	Enable for oblique or aerial scenarios and geometrical projection.
Illumination and source spectrum	KEEP	Mandatory model branch for visible/NIR/SWIR radiometric predictions.
Atmospheric extinction	KEEP	Fast-mode coefficient with stated units and spectral validity.
Atmospheric spectral transmission	KEEP	Import wavelength- and range-dependent values when available.
Path radiance	KEEP	Model or import independently from direct-path transmittance.
Turbulence MTF / Cn^2	KEEP	Use scenario-dependent defaults, with expert controls.
Residual stabilization / image motion	KEEP	Input measured angular jitter or time-series data.
Lab vs field / video vs still	KEEP	Explicit toggles; do not silently mix unlike conditions.
Standard DRI / task thresholds	KEEP	Retain established, clearly labeled criteria; no adaptive criteria.
MTF / AMOP / MTDP / MDSP curves	KEEP	Primary engineering outputs; concise summaries on dashboard.
MCP/MRC data option	ADVANCED	Measured-system performance route when available.
NIIRS / aerial interpretability	ADVANCED	Separate mode only if an aerial-image task needs it.
Legacy hardware control dialogs	REMOVE	Do not replicate TRM4 dialog proliferation in the modern UI.

TRM4 source: Sections 2.19-2.28 (pp. 77-102), Sections 3.11-3.13 (pp. 142-159). Standard target assumptions are scenario-specific; see Table 2.1 (p. 87).

05 Proposed application interface

Default: Quick Prediction

Present only the fields required to define the camera, scene and task. Additional parameters should come from profiles and transparent presets.

ON-SCREEN INPUT	BEHAVIOR / DEFAULT
Camera profile	Select sensor + optical configuration; allow manual profile creation.
Band / zoom / focal length	Prepopulated from profile; visible if multiple channels/zooms exist.
Target preset and orientation	Vehicle/person/custom; dimensions always reviewable.
Target-background signature	Temperature contrast for thermal; reflective-band contrast data as appropriate.
Atmosphere	Clear / average / adverse preset or user-entered validated extinction.
Task & observation	Standard detection / recognition / identification, lab or field, still or video.

Three logically grouped screens

SCREEN	VISIBLE CONTENT	ADVANCED DRAWER
Setup	Camera + optics, target, environment and task.	Sensor noise, spectral curves, 3D geometry.
Results	DRI range, target pixels, resolving cycles, performance summary.	Directional MTF/AMOP, MTDP/MCP, SNR and budgets.
Validation	Upload calibration results, compare measured and predicted.	Raw frames, test metadata, diagnostics, uncertainty assumptions.

Automatic calculations (read-only by default)

Horizontal/vertical FOV and IFOV; Nyquist frequency; effective aperture and F-number; scene radiometric quantities; selected MTF chain; spatial sampling; task threshold mapping; transmission and path-radiance quantities; range performance.

UX principle: Editing an independent variable should update dependent results. Inconsistent redundant inputs should generate a warning, not silently override the physical calculation. The application must show which value came from a manufacturer, measurement, import, preset or assumption.

06 Optics and file-import improvements

A. Optical response data precedence

- **Mode A: Computed optics.** Use geometry, wavelength, diffraction and an optional aberration model.
- **Mode B: Imported optical MTF.** A measured or simulated complete optics MTF replaces computed optics MTF; do not apply diffraction a second time.
- **Mode C: Imported device prefilter MTF.** If the measured data cover optics plus detector and pre-sampling effects, bypass those submodels to avoid double counting.
- **Device vs system response.** Distinguish front-optics MTF, prefilter MTF, delivered-image SFR and overall observer response.

B. Import contracts (recommended)

FILE / DATA	REQUIRED VALIDATION
Optics MTF	Spatial-frequency units (cycles/mm or cycles/mrad), tangential/sagittal or H/V axis, wavelength or spectral weighting, field position, zoom/focus, MTF range [0,1].
Transmission	Wavelength units, absolute vs normalized values, optical surfaces covered, and valid spectral band. Never multiply a full-system throughput twice.
Detector QE	Wavelength range, fractions vs percentages, sensor configuration and sensor temperature where relevant.
Calibration metadata	Instrument, date, method, focus, temperature, frame processing, uncertainty and configuration identifier.

C. Important physical constraints

- Keep **horizontal and vertical** MTF information when possible. A single scalar cannot describe anisotropic optics or stabilization.
- Validate optical frequency coordinates before resampling: focal-plane cycles/mm are not the same as object-space cycles/mrad.
- Use a properly sampled optical frequency grid; do not infer complete optical quality from MTF50 alone.
- Model aliasing/undersampling explicitly where needed. TRM4 uses **AMOP**, not a naive total-MTF product, for its bar-target performance calculations.

Source mapping: TRM4 Sections 2.6.5-2.6.9 (pp. 31-34), 2.23 (p. 83), 4.1-4.3 (pp. 161-180). ISO 12233:2024 is relevant for measured camera spatial-frequency response.

07 Detector and image-chain modeling

Branch the model by actual sensor physics

CAMERA CLASS	CORE MODEL	MEASUREMENT SHORTCUT
Day / NIR / SWIR	Spectral radiance + optics/QE + integration time + electrons + noise + saturation.	System-level measured sensitivity or MCP/MRC if suitable.
Cooled MWIR	Band-weighted scene radiance; dark/read/shot noise; temperature dependence.	Measured NETD/MTDP or calibrated detector performance.
Uncooled LWIR	Thermal responsivity and radiometric/noise response, with background dependence.	Measured NETD + MTDP when available.

Changes to the calculation and processing engine

- **Data-source switches:** each module must identify whether a value is modeled, measured, or imported, and must not combine mutually exclusive paths.
- **Noise separation:** model temporal noise, residual fixed-pattern/non-uniformity noise, quantization and shot noise only where physically applicable.
- **Saturation and dynamic range:** check full-well/ADC clipping and note when linear signal-noise assumptions are invalid.
- **Processing controls:** represent real denoising, frame averaging and sharpening through measured response or explicit processing parameters; document tradeoffs.
- **Temperature dependence:** store the conditions under which NETD or detector data were measured; flag extrapolation.
- **Video:** distinguish integration time from frame interval and avoid assuming that frame rate alone defines sensitivity.

Laboratory support to add

IMPORT	USE IN THE APPLICATION
Calibrated test videos	Temporal noise / fixed pattern separation, NETD estimates, pixel statistics.
Flat fields and dark frames	NUC/non-uniformity and offset checks.
Slanted-edge / bar patterns	Measured delivered-image SFR, MTF or contrast response.
Camera metadata	Automatic matching of test data to actual sensor configuration.

Source mapping: TRM4 Sections 2.10-2.13 (pp. 45-65); Sections 3.8-3.11 (pp. 131-152). EMVA 1288 and ISO 12233 provide complementary measurement frameworks for digital cameras.

08 Atmosphere, turbulence & stabilization

Atmospheric model: two fidelity levels

FAST ENGINEERING MODE	DETAILED SPECTRAL MODE
Beer-Lambert effective extinction coefficient; documented assumption and wavelength validity.	Wavelength/range-dependent transmission; imported MODTRAN data or equivalent calculated spectrum.
Preset environmental cases and immediate comparison.	Atmospheric profile, temperature, humidity, aerosols, pressure and line-of-sight geometry.
Simplified radiance / contrast impact with warnings.	Import path radiance separately and integrate over sensor spectral response.

Do not equate visual visibility with a universal infrared extinction coefficient. TRM4 restricts its visibility-based approach to its visible category and separately permits external spectral data. Keep atmospheric transmittance and path radiance distinct: both may influence signal contrast, but they represent different physical contributions.

Turbulence

- Use the camera-to-target geometry and a selected/estimated turbulence profile, not a constant MTF for all distances.
- Preserve short- versus long-exposure assumptions where used; turbulence treatment must match the actual acquisition/processing regime.
- Display the turbulence component and its contribution to the limiting spatial frequencies.

Stabilization

- Accept measured image motion in pixels, microradians, or time-series samples; store axis conventions and sample rate.
- Distinguish residual random jitter, intra-frame motion blur, vibration, slow drift and pointing offsets.
- Calculate a justified blur/MTF effect; optionally provide PSD-based diagnostics for vibration-sensitive installations.

Source mapping: TRM4 Sections 2.19 (pp. 77-78), 2.27-2.28 (pp. 96-102), 4.2.10 and 4.2.13 (pp. 175-178). External data providers: MODTRAN and HITRAN.

09 Results, diagnostics & uncertainty

Always-visible result dashboard

OUTPUT	WHY IT MUST BE AVAILABLE
Standard D / R / I range	Primary, task-specific modeled result with named target/scenario assumptions.
Projected target pixels	Fast visual explanation of geometric and sampling limits.
Resolved cycles / AMOP evidence	Link between target pattern, angular sampling and modeled visibility.
MTF50 and MTF at Nyquist	Compact diagnostic summaries; not substitutes for the full model.
CNR or system contrast metric	Explains image separability for compatible measurement conditions.
Dominant limiting factor	Sensitivity sweep to attribute changes to optics, atmosphere, noise, etc.
Assumptions & confidence	Data provenance, bounds and uncertainty/unsupported-condition warnings.

Recommended engineering views

Full curves: optical/device/system MTF or AMOP, MTDP/MDSP or MCP, signal/noise vs range, effective transmission, and DRI-versus-parameter sensitivity. Keep these in an expandable panel rather than showing every graph by default.

Diagnostics the application should generate automatically

- Conflicting entered and calculated FOV; impossible optical geometry or inconsistent F-number.
- MTF outside [0,1], unspecified axis, missing frequencies near Nyquist, or incompatible spectral coverage.
- Mixed absolute/normalized transmission, throughput counted twice, or transmission outside [0,1].
- Unsupported sensor-type assumptions; low-SNR results; saturation; questionable NETD test conditions.
- Comparison made between lab and field conditions without the same target, scene and measurement configuration.

Confidence is not a magic score: when input uncertainties are known, use a sensitivity or Monte Carlo study and report credible input ranges. If they are unknown, label the output as a nominal prediction rather than providing a fictitious precision.

10 Verification and acceptance plan

Before adding new features, compare the application against known physical identities, laboratory measurements, and controlled TRM reference cases. Maintain a regression suite as each processing path is introduced.

TEST AREA	MINIMUM ACCEPTANCE CHECK
Geometry	FOV/IFOV/target-pixel values reproduce independent calculations; angle and unit conversions match.
Sampling	Nyquist values correct for pitch and focal length; binned modes update effective sampling.
Optics	Known diffraction-limited curve is reproduced; imported optical MTF is not re-diffracted.
Spectral response	Correct overlap and weighting of detector QE, optics, filter and atmospheric spectra.
Transmission	Absolute vs normalized CSV tests; Beer-Lambert behavior for specified effective extinction.
Radiometry/noise	Known synthetic signals reproduce shot/read/dark noise and clipping behavior.
Thermal	Measured NETD/MTDP inputs yield reproducible predictions under the documented scene.
Turbulence/motion	Zero motion has no added blur; nonzero jitter reduces relevant detail as expected.
Measured MTF	Cross-check camera SFR against a calibrated reference target and protocol.
DRI tasks	Use standard task/preset definitions; reproduce chosen reference calculations within stated tolerance.
Edge cases	No target contrast, very short range, invalid extrapolation, missing data and unit errors are handled clearly.
Provenance	Every result records model version, import hashes, units, sensor configuration and assumptions.

Recommended validation artifacts

Maintain a controlled dataset of optics CSV files, spectral transmittance files, camera-characterization runs, target scenarios and reference outputs. For each test case, record expected behavior, numerical tolerance, source and reviewer.

> **Quality gate:** do not release a new modeled pathway until (1) numerical unit tests pass, (2) independent reference comparison is documented, and (3) the UI shows any applicable range or validity limitations.

11 Prioritized implementation roadmap

PRIORITY	DELIVERABLE	DONE WHEN
P0 / Foundation	Camera profiles, simplified setup, input validation, optics/detector models, baseline DRI and core plots.	Reference-camera and target scenarios are repeatable and internally consistent.
P1 / Measured data	Zemax CSV import, spectral transmission, measured NETD/MTF, provenance and error checks.	Imported and computed paths can be compared without double counting.
P1 / Environment	Fast atmosphere, spectral import, path radiance, turbulence and measured jitter.	Environmental sensitivity is visible, physically labeled, and tested.
P2 / Engineering UX	Bottleneck analysis, uncertainty view, lab-data tools, video/image diagnostics.	Analyst can explain changes to DRI and trace inputs to measurements.
Optional	NIIRS, 3D target orientation, legacy scanners/intensifiers.	Only implemented when the required application class exists.

Explicitly outside the present change package

No new Johnson criterion; no adaptive task thresholds; no learning from observer feedback; no change to historical or standard DRI criteria. Conventional TRM task criteria remain an explicitly selected modeling input. Real-image evaluation, if implemented, is limited to calibrated measurements and diagnostic comparisons rather than learning new perception thresholds.

Reference basis and further technical reading

Primary source: Fraunhofer IOSB, TRM4.v3: Parameter Description and Model Documentation, Technical Report 2021/04, especially Chapters 2-4 (pp. 17-180). Supplied by the user.

ISO 12233:2024: Digital cameras - Resolution and spatial frequency responses. https://www.iso.org/standard/88626.html

EMVA 1288, Release 4.0: Objective characterization of industrial cameras. https://www.emva.org/standards-technology/emva-1288/

MODTRAN: Radiative transfer software for LOS spectral transmission and radiance. https://www.spectral.com/our-software/modtran/

HITRAN: Molecular spectroscopic database. https://hitran.org/

Interpretation: The TRM4 references identify existing model capabilities. The KEEP / SIMPLIFY / ADVANCED / REMOVE dispositions, UI structure, file-contract checks and roadmap are proposed engineering decisions, not requirements stated by Fraunhofer.